

[Technical Paper]

Low Warpage Coreless Substrate for IC Packages

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Abstract

Coreless substrate is excellent for fine patterning, small via pitches, and transmission property, and it is a promising IC packaging method for the next generation. Warpage of coreless substrate is generally large compared to the other types of IC packaging substrates because of inadequate rigidity, so the most important problem for the application of coreless substrates for high-end BGAs is warpage reduction during a reflow process. So far, only a limited number of reports have been focused on coreless substrates for large size IC packages. Moreover, very few examples have discussed substrate layer structural designs for warpage reduction and reliability improvement in IC assembly processes. In our study, we focused on the development of coreless substrates for large size ICs. To achieve our goal, we adopted the following development procedure. First, we designed analytical models with different layer structures composed of two kinds of insulating materials and estimated the effective layer structures for warpage reduction by numerical analysis. Next, we prepared the real coreless substrates with the same structure as the analytical models and evaluated their actual thermal behavior. Finally, we investigated the thermal stress reliability of IC mounted substrates. As results of these examinations, we successfully developed low warpage and high reliable coreless substrate by introducing high rigidity materials only in the external layers of the substrate.

Keywords: Coreless substrate, Low warpage, IC mounted reliability, Layer structure, Prepreg

1. Background

With a continuous enhancement of performance and feature downsizing of electronic products, the use of Ball grid array (BGA) packages is spreading rapidly as electronic products because of their advantage in higher pin counts. Moreover, in keeping with the IC upsizing pace, even more fine BGAs are necessary, and as a result, the development of high reliability assembly processes for fine-pitch ICs had become essential. We believe that one of the important factors for the BGA performance is dielectric materials, which include ceramics[1] and organic materials.[2–4] We have been continuously developing buildup BGA substrates making use of organic materials[5] that are more advantageous over ceramics because of their low cost and microfabrication compatibility.

Buildup substrate consists of core layers reinforced by glass cloths and buildup layers consisting of resin films. Generally, the thickness of core layers is in a range of 200–

800 μm . Therefore, via plated through holes (PTH), which penetrates through core layers, is needed for interconnection. The minimum limit of PTH pitch is estimated to be 200 μm , but solder bump pitch for ASICs is expected to become around 100 μm . [6] Therefore, the development of finer interconnection technologies for organic BGA substrates is highly essential. Whereas vias of buildup layers can be made by a laser process, which is suitable for microfabrication and the minimum via size is estimated to be 30 μm . Therefore, high density wiring can be realized by the adoption of coreless substrate, which doesn't include core layers. [7, 8]

The comparison of properties between coreless and buildup substrate is shown in Fig. 1. In general, the adoption of coreless substrates for IC packages has three advantages, including high wiring design flexibility owing to fine via pitch, power source improvement because of low impedance, and large signal integrity. However, core-

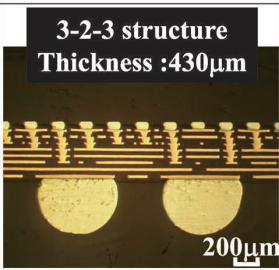
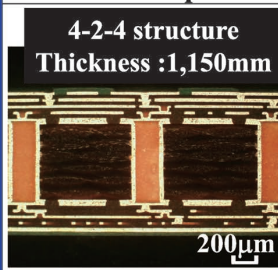
Properties	Coreless	Bulidup
Cross-sectional view of structure		
Patterning scale	Fine	Rough (Core layer)
Via pitch	Small	Large (PTH)
Transmission	Excellent	Poor (Core layer)
Warpage	Large (Low rigidity)	Small

Fig. 1 Comparison of properties between coreless and buildup substrate.

less substrates include no core layers with high rigidity, thermal warpage of coreless substrates becomes larger than that of the buildup substrates. As a result, large ICs cannot be mounted on coreless substrates. The reliability issue due to large warpage has prevented the widespread use of coreless substrates including high end use. To date, coreless substrates have been used in limited applications such as small size of CSPs. In this study, we aim to develop coreless substrates with very low warpage below 100 μm for a BGA bump pitch of 1 mm, on which large size of ICs greater than 20 mm squares can be mounted. Then we performed additional verification experiments and added the assumed mechanisms for the results to our previous report.[9]

2. Experimental Procedures

2.1 Preparation of coreless substrates

First, we researched the warpage improvement methods of coreless substrates. To improve the rigidity of coreless substrate, the use of prepregs as a dielectric material has been recently reported.[10] Prepreg is adhesion resin sheet that is reinforced by glass cloths, and this material is widely used as a bonding sheet. However, prepregs could cause skew in differential wirings for high-speed transmissions due to the presence of glass cloth, and an explanation figure of skew generation mechanism is shown in Fig. 2. Therefore, coexisting of low warpage and excellent transmission property is very important and it can be achieved by using a minimum number of prepregs. In this study, we examined to reduce warpage by the application of two kinds of dielectric materials, which are prepreg and resin. To control the warpage from the moment effect, we arranged prepregs in external layers. Moreover, for the

skewless transmission, we arranged resins in internal layers. Thus, we propose coreless structures with external prepregs and internal resins to achieve both properties.

Then we prepared four kinds of coreless substrates with different layer structures. Figure 3 describes the wiring layer structure of the prepared coreless substrate. In this figure, white area indicates the wiring layer and the other area is insulating layer. The Cu ratio of seven layers was roughly adjusted to be equivalent for warpage reduction. The coreless substrates had an area of 42.5 mm squares and thickness of 0.4 mm. We also included IC bumps and daisy-chain patterns for evaluation of IC and BGA bump interconnection reliability with 20 mm squares' ICs. Figure 4 shows the external appearance of coreless substrates.

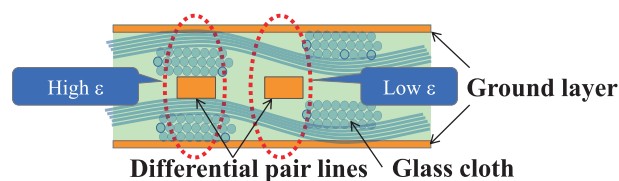


Fig. 2 Skew problem of differential pair lines in prepreg.

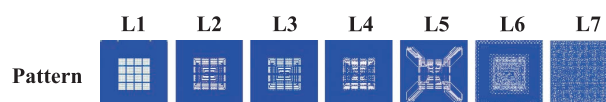


Fig. 3 Wiring layer structure of prepared coreless substrate.

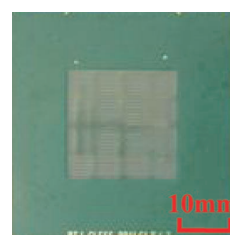


Fig. 4 Appearance of coreless substrate.

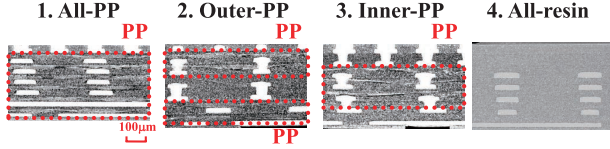


Fig. 5 Cross-sectional images of substrates.

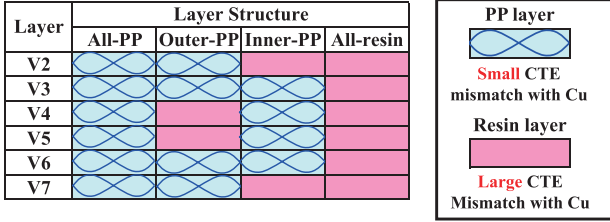


Fig. 6 Schematic figure of layer structures.

The layer structures of the substrates are shown as follows:

All-PP: all prepreg layers

Outer-PP: two internal resin layers / four external prepreg layers

Inner-PP: two external resin layers / four internal prepreg layers

All-resin: all resin layers

The cross-sectional images of substrates and schematic figure of layer structures are shown in Fig. 5 and Fig. 6. Here, only the layer composition was changed by different combination of two buildup materials named GX-13 (resin) and GX13-GC (prepreg, GX-13 resin impregnated with glass cloths) commercialized by Ajinomoto Fine-Techno Co., Inc.

2.3 Evaluation methods — Numerical analysis —

Laminated shell models were created with great care to describe the wiring and vias of all layers of substrates. They were converted from EDA-CAD by the BoardWARP system, which we previously developed. With this system, we can obtain precise mesh data from original CAD data. The mechanism of this system has previously been described in detail.[11] This time all the wiring layer images of the CAD and analytical model data were produced by our method. The analytical parameters were mesh pitch = 50 μm , total number of elements = 810,000, and computing time = 50 h. Images of the analytical model are shown in Fig. 7. In this figure, white area indicates the wiring layer and the other area is insulating layer. By comparing both, we confirmed CAD data were faithfully converted.

We executed thermal warpage analyses with the general structural analysis software ABAQUS6.8-1 as an analytical solver. This time, we performed viscoelastic analyses for accurate prediction. The theory has been described in

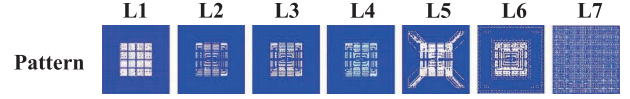


Fig. 7 Wiring layer structure for numerical analysis.

detail to various technological theses.[12] Here, the viscoelastic theory used for viscoelastic analysis is concisely described. In viscoelastic analysis, it is necessary to use the relaxation modulus that is the function of time (t). The viscoelastic equation, which is the relation between the stress σ and the deformation ε , can be expressed below with $G(t)$.

$$\sigma(t) = \int_{-\infty}^t G(t-\tau) \frac{d\varepsilon(\tau)}{d\tau} d\tau \quad \dots(1)$$

The thermal viscoelastic equation can be expressed below with relaxation modulus $G(e, T_0)$.

$$\sigma(t) = \int_0^t G(e-\tau, T_0) \frac{d\varepsilon(\tau)}{d\tau} d\tau \quad \dots(2)$$

Here, e is pseudo time defined by the next equation. T_0 is a reference temperature in time-temperature shift parameter a_T .

$$e = \int \frac{dt}{a_T [T(x, t)]} \quad \dots(3)$$

To obtain the relaxation modulus (master curve), we executed the dynamic frequency dispersion examination. The Williams-Landel-Ferry law (WLF Law) was used for deriving a_T in the master curve. The frequency dispersion curves at different temperature were overlapped with the curve of T_0 , and the master curve was obtained. The relaxation modulus is expressible by the use of the prony series shown in the next expression.

$$G(t) = G_0 + \sum_k G_k \exp(-t / \tau_k) \quad \dots(4)$$

Finally the viscoelasticity constant G_k and τ_k can be calculated by curve fitting of the acquired master curve.

As well as the faithful analytical model, precise material property is the essential factor influencing the analytical accuracy. All coreless substrates consisted of resin, prepreg, solder resist, and copper wirings and all material properties for the analyses were evaluated initially. The buildup materials and solder resist were defined as viscoelastic materials, and copper was defined as an elastic material. The material properties of Cu were described in Table 1. For evaluation of viscoelastic properties, we used Rheometrics RSA-II. The measurement was executed every 5°C. 3 points bending mode was adopted for prepreg, and tensile mode was adopted for resin and solder resist. Master curve and shift parameter data were acquired

Table 1 Material properties of copper.

Material	Modulus (GPa)			Poisson Ratio			CTE (1/°C)
	Ex	Ey	Ez	vxy	vyz	vxz	
Cu	55.9			0.3			1.63E-05

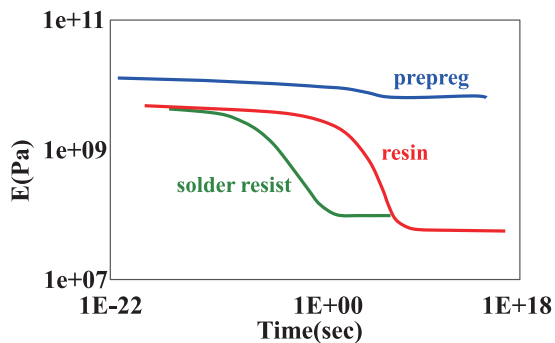


Fig. 8 Master curves of materials.

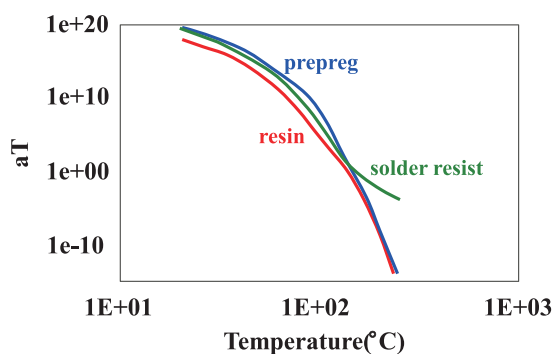


Fig. 9 Shift parameters of materials.

by smoothly conjugating all of the frequency dispersion data. Evaluation results of master curves and shift parameters are shown in Fig. 8 and Fig. 9. CTE properties were measured by using a Seiko Instruments TMA/SS6100. CTE measurement with compressive mode was adopted for prepreg, and tensile mode was adopted for resin and solder resist. CTE data on every 5°C defined as the initial temperature at -50°C are shown in Fig. 10. In these figures, the anisotropy of the prepreg was also measured, but in the laminated shell element, the anisotropy was not considered. Therefore, the CTE difference in the thickness direction was ignored because the thermal expansion difference and flexural elasticity in the surface direction dominate the substrate warpage behaviors. The three nodes of the substrate corner positions were fixed in the vertical (z) direction, and the single node of the center was fixed in all directions and rotations.

With these conditions, we performed the viscoelastic analyses. In this study, we paid especially attention of the relative warpage at 220°C based on the initial warpage because of the melting temperature of SAC solder.

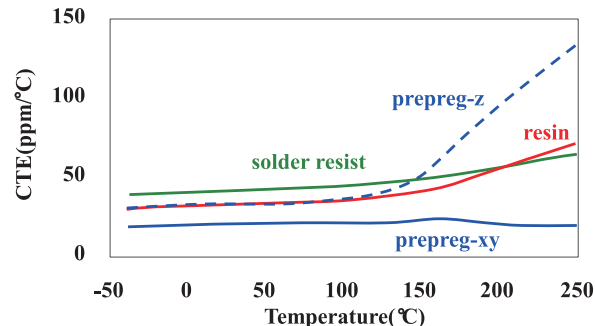


Fig. 10 CTE data of materials.

2.2 Evaluation methods — Thermal warpage measurement —

The shadow moiré method[13] is widely used for measuring the temperature-dependent warpage of various electronic products, including IC packages. Moreover, this technique has been authorized as a method for temperature-dependent warpage measurement in JEDEC[14] and JEITA[15] standards. For these reasons, we adopted this method to evaluate the temperature-dependent warpage with the TherMoiré AXP systems commercialized by AkroMetrix, LLC. In the shadow moiré method, the sample is heated from only the bottom side. Therefore, the temperature difference between both sides of the sample is generated when the same heating profile as a general reflow process is applied. The heating speed must be carefully controlled by considering the thermal conductivity of the sample, because the temperature difference causes an measurement error.

The conditions of the temperature-dependent warpage measurement are as follows:

- Substrate support: quartz plate (with IC surface up)
- Temperature controlling method: thermocouples attached to both sides of the reference substrate, which is placed next to the measurement sample for temperature control.
- Temperature range: 30 to 230°C
- The warpage data for each temperature were acquired by measuring the relative warpage based on the warpage at the room temperature.

2.3 Evaluation methods — IC assembly test —

To evaluate the acceleration reliability of solder joints we performed LSI assembly test with LSI mounted pack-

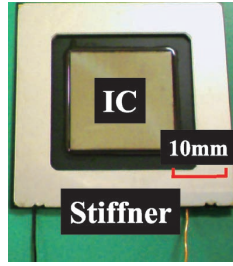


Fig. 11 External view of the IC-mounted coreless substrate.

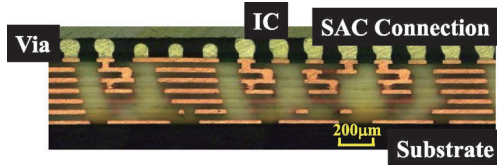


Fig. 12 Cross-sectional view of the IC-mounted coreless substrate.

Table 2 Test conditions of the IC reliability test.

Thermal Cycle	-25/+125 (°C) 1,000 cycles
IC Size	20 mm square, 0.15 mm thick
Substrate Size	42.5 mm square, 0.3 mm thick

ages. Twenty test pieces were prepared for each kind of IC-mounted coreless substrate, and gap between IC and substrate was filled with a commercialized underfill. The mounted IC parameters are as follows: area, 20 mm squares; thickness, 0.15 mm; solder bump, SAC; and including interconnect pattern-staggered daisy chain vias.

The external and the cross-sectional view of the IC-mounted coreless substrate are shown in Fig. 11 and Fig. 12, respectively. The reliability test conditions are summarized in Table 2, and the passing requirements are shown as follows.

1. Resistance increase should be less than 100%.
2. Failure rate should be 0%.

3. Results

3.1 Numerical analysis

Results of numerical analysis are shown in Fig. 13. We calculated the warpage at 220°C relative to that of the initial temperature. All the substrates were estimated to deform into a convex curvature to the IC-mounted side, and the amount of warpage was 111 μm for All-PP, 120 μm for Outer-PP, 179 μm for Inner-PP, and 269 μm for All-resin. These values agree with our expectation that Outer-PP should have almost the same warpage as All-PP.

3.2 Warpage measurement

Figure 14 shows warpage measurement results at 220°C

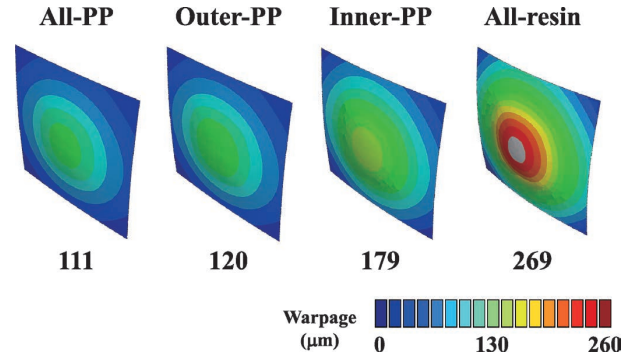


Fig. 13 Warpage simulation results.

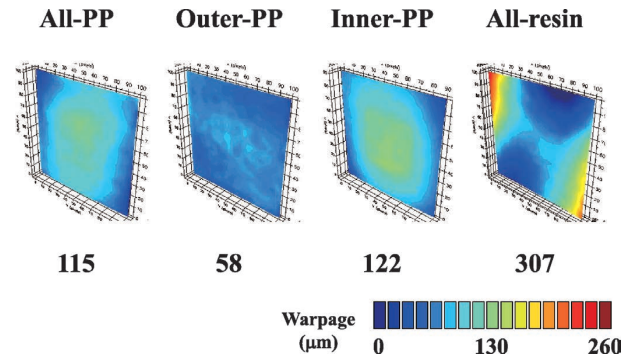


Fig. 14 Warpage measurement results.

relative to that of the initial temperature. Except for All-resin, all other substrates deformed into a convex curvature. All-resin deformed into a twisted shape, and hence, the sign of the warpage was undefined. The amount of the warpage was measured to be 115 μm for All-PP, 58 μm for Outer-PP, 122 μm for Inner-PP, and 307 μm for All-resin; Outer-PP exhibited the minimum amount of warpage. As a result, we succeeded in reducing the warpage by the introducing relatively high-rigidity materials in the external layers of the substrate.

3.3 IC assembly reliability test

The IC assembly test results are given in Table 3. The All-PP and Outer-PP substrates exhibited no failures during the thermal cycle test. Obviously, they have the desired design quality and high reliability in their solder joints. However, as was the case with the All-resin structure, the Inner-PP structure was found to exhibit failures

Table 3 IC assembly test results.

Sample	IC assembly test results	Warpage (μm)
All-PP	Passed	115
Outer-PP	Passed	58
Inner-PP	Failed	122
All-resin	Failed	307

in spite of a small warpage in the reflow temperature.

4. Discussion

Our results are discussed from four perspectives. The first discussion theme is the reason for the Inner-PP structure failure. Warpage change during the cooling process is an important factor when we discuss assembly reliability. Figure 15 shows the relative warpage changes during the heating process. As mentioned before, the Inner-PP structure exhibited a small warpage increase upon heating. Additionally, the relative warpage changes during the cooling process from 220°C are shown in Fig. 16. In contrast to heating process, the Inner-PP structure exhibited a large warpage increase, which is over twice of the All-PP and Outer-PP structure. This means that the warpage of the Inner-PP and All-resin structures greatly increased after solder ball solidification. Therefore, huge stresses were expected to be generated in the solder joints of these structures, rendering it difficult to ensure assembly reliability. This is the major reason for the decrease in reliability of the Inner-PP structure. As a result, this means that the effective factors for high reliability are not only low warpage at the solder melting point but also small temperature-dependent warpage during cooling process.

The second discussion theme is the reason for the warpage increase of Inner-PP structure during cooling process.

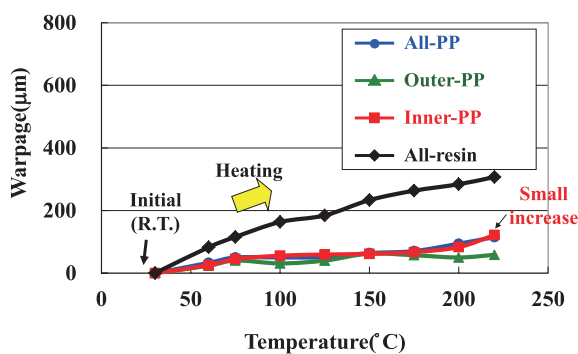


Fig. 15 Relative warpage changes upon heating.

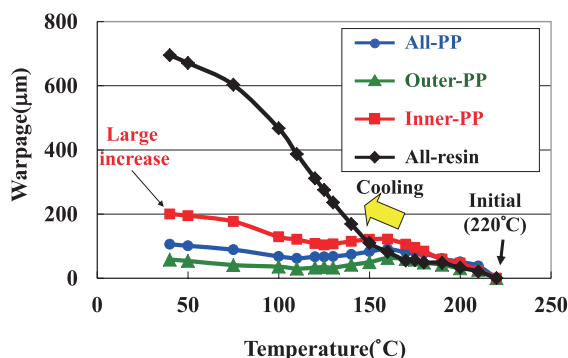


Fig. 16 Relative warpage changes upon cooling.

Figure 17 shows the thermal cycle behavior of prepreg and resin from room temperature to 220°C. In this figure, prepreg returned to their original length after thermal cycle, whereas resin shrunk drastically. This is because resin has no restraint in the surface direction by glass cloth. Therefore, the effective CTE value of resin during cooling process became larger. As a result, the warpage in cooling process became larger especially in external resin structures.

In third, we discuss the reason why Outer-PP structure exhibited the smallest warpage. Figure 18 shows the layer structures of Outer-PP and All-PP structures. The CTE values are matched with Cu in the outermost layer of both structures. Additionally, for Outer-PP structure, the stress in surface layer is absorbed by low modulus material arranged in internal layer. We conclude that the stress relaxation effect in resin layer is a major reason for warpage decrease from our numerical analysis results. The stress is thought to be released without being accumulated as tiredness, this can be expected by the reliability test result of Outer-PP structure.

Finally, the results of numerical analysis and measurement should be compared. We can predict the warpage of All-PP structure in high accuracy though the prediction accuracy of Outer-PP structure is poor. In case of the structure composed by plural dielectric materials, which are Outer-PP and Inner-PP structure, the measurement value has become smaller than our simulated value. We concluded that the stress relaxation effect had been generated in resin layer. In our analysis, this effect had not been

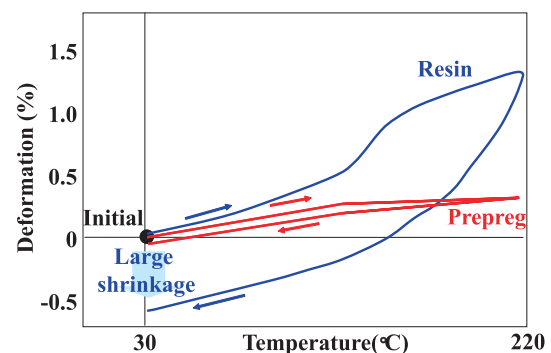


Fig. 17 Thermal cycle behaviour of prepreg and resin.

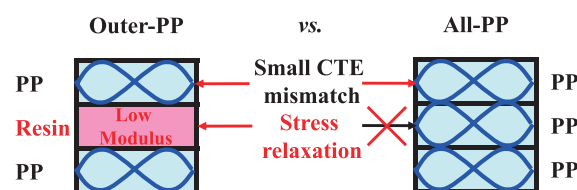


Fig. 18 Layer structures of outer-PP and All-PP.

considered, so the consideration is our next target for more accurate warpage prediction.

5. Conclusions

For the application of coreless substrate to IC package, we investigated the effect of layer structure on warpage reduction of coreless substrate consisted of resins and prepregs as dielectric materials by checking with numerical analysis and warpage measurement technics. Moreover, we conducted IC assembly reliability tests with real coreless substrates. As a result, we derived the following conclusions.

1. The most effective structure to reduce warpage in our study was the application of only prepreg materials in both external layers. The warpage gradually increased with the dielectric combination in the following order: all prepreg < inner prepreg layers < all resin.
2. Temperature-dependent warpage behavior is very important for IC assembly reliability alongside the warpage at the melting point of solder joints. We confirmed that the IC-mounted substrate with the effective layer structures cleared 1,000 thermal test cycles.

In conclusion, we successfully developed a low warpage coreless substrate (58 μm) that can be applied to 20 mm squares of ICs by introducing prepregs only in both external layers of the substrate.

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